

IN THE PATENT OFFICE, INDIA

Under The Patents Act, 1970 (as amended)

COMPLETE SPECIFICATION

(Under Section 10 of The Patents Act, 1970)

A SYSTEM AND METHOD FOR CONTEXT-ADAPTIVE CHILD DAILY ROUTINE GENERATION WITH REAL-TIME AMBIENT CONDITION INTEGRATION AND CAREGIVER-AWARE PERSONALISATION

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FILING OFFICE: Indian Patent Office (preferably Mumbai / Delhi
branch)

PRIORITY DATE: [Date of Filing]

This document has been prepared as a technical specification for patent prosecution.

It must be reviewed, verified, and filed by a registered Indian Patent Agent.

CONFIDENTIAL — PRIVILEGED DOCUMENT

Prepared: May 2025 | Product: AmyNest AI Routine Planner

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ABSTRACT

The present invention discloses a computer-implemented system and method for generating personalised daily schedules for children by integrating real-time geolocation-derived ambient weather data, developmental age-band classification, caregiver identity context, and regional dietary preferences through a hybrid AI-and-rule-based scheduling engine.

A key aspect of the invention is a **silent pre-generation ambient condition detection module** that, immediately before a user-triggered schedule generation event, performs a geolocation query and fetches meteorological data from an external weather API (Open-Meteo) to derive an outdoor suitability verdict (*suitable / limited / unsuitable*). The detection module employs a single shared in-flight promise to prevent duplicate permission prompts under concurrent invocations, and includes an explicit user-preference preservation mechanism that discards a late-resolving detection if the user has explicitly set a weather preference during the detection window.

The generation engine classifies children into five developmental age bands (Infant 0–11 months, Toddler 1–3 years, Preschool 3–5 years, Early School 5–10 years, Pre-Teen 10–15 years) and applies age-specific activity vocabularies, meal guidance, and sleep requirements. A caregiver-adaptive post-processor adjusts schedule instructions based on who is caring for the child (mother, father, both parents, grandparent, or babysitter). A fridge-item-aware, region-specific, allergy-safe meal enrichment pipeline produces exactly four distinct meal options per slot using generative AI. A child energy-curve engine heuristically re-orders activities to align demanding cognitive tasks with detected peak-focus windows and rest/sensory tasks with low-energy windows.

The invention produces schedules on cross-platform client applications (web browser and native mobile device) and is particularly suited to the Indian parenting context through its support for regional cuisines, dietary restrictions, and tri-lingual content delivery.

[Abstract must not exceed 150 words in the filed version — the above is the full technical abstract for specification purposes. A condensed 150-word version for Form 2 follows the Claims section.]

1. FIELD OF THE INVENTION

The present invention relates to computer-implemented systems and methods for generating personalised child daily schedules. More particularly, it relates to a hybrid artificial intelligence and rule-based routine generation engine that silently integrates real-time device geolocation, ambient weather data, developmental age-band parameters, caregiver context, and regional dietary preferences to produce adaptive, time-anchored daily schedules for children, delivered across web-browser and native mobile-device platforms.

2. BACKGROUND OF THE INVENTION

The management of a child's daily routine is a cognitively demanding task for parents and caregivers. Existing digital tools — including general-purpose calendar applications and static parenting schedule templates — suffer from the following limitations:

1. **Static weather handling:** Existing systems require the parent to manually observe and enter outdoor weather suitability before generating a schedule. When the parent omits this step, the system defaults to a fixed value (typically "outdoor activities enabled"), which results in schedule recommendations that are inappropriate for the actual ambient conditions (e.g., an outdoor park visit is recommended on a day with heavy rain or extreme heat).
2. **Age-agnostic scheduling:** Prior art systems provide uniform activity templates without differentiating between the developmental needs of infants, toddlers, preschool-age children, school-age children, and pre-teenagers. Activities such as "Homework — Pomodoro technique" are developmentally inappropriate for a two-year-old, while "Tummy Time" is inappropriate for a ten-year-old.
3. **Caregiver-oblivious instructions:** Generated schedule instructions do not adapt to whether a grandparent, a babysitter, or a parent is in charge. Instructions that assume a parent's judgment (e.g., "cook a complex meal from scratch") are inappropriate when communicated to a babysitter.
4. **Generic meal recommendations:** Existing systems recommend generic meal names without considering the family's available ingredients (fridge inventory), regional Indian cuisine preferences (North Indian, South Indian, Bengali, etc.), or specific dietary restrictions (Vegan, Jain, etc.).

5. **Static activity ordering:** Generated schedules place activities in a fixed temporal order, without adapting to a child's historically observed energy curve (peak-focus windows, low-energy windows, calm pre-bedtime windows), leading to sub-optimal learning and rest outcomes.

There is therefore a need in the art for a system and method that (a) silently, automatically integrates real-time ambient weather data before schedule generation without disrupting the user's workflow or overriding an explicit user choice; (b) applies developmental age-band intelligence; (c) adapts instructions to the active caregiver; (d) produces region-specific, fridge-aware meal recommendations; and (e) heuristically re-orders activities according to the child's observed energy profile.

3. OBJECTS OF THE INVENTION

1. To provide a silent, non-blocking, pre-generation ambient weather detection mechanism that derives an outdoor suitability verdict from real-time geolocation and meteorological API data.
2. To ensure the pre-detection mechanism is idempotent and concurrent-safe, such that multiple simultaneous invocations share a single geolocation request without generating duplicate permission prompts.
3. To preserve explicit user weather preferences so that a late-resolving ambient detection result cannot override a preference manually set by the user during the detection window.
4. To classify children into developmental age bands and apply age-appropriate activity vocabularies, meal guidance, and sleep parameters.
5. To adapt generated schedule instructions based on the identity and capability profile of the active caregiver.
6. To produce region-specific, fridge-item-aware, allergy-safe meal recommendations using a generative AI enrichment pipeline.
7. To heuristically re-order schedule activities according to a child's historically observed energy profile.
8. To enforce school-block constraints on AI-generated schedules so that activities are never placed during the child's school hours.

4. SUMMARY OF THE INVENTION

According to a first aspect, the present invention provides a computer-implemented **system** for generating a personalised daily schedule for a child, comprising: a pre-generation ambient condition detection module; an age-band classification engine; a

hybrid schedule generation engine comprising a generative AI path and a deterministic rule-based path; a caregiver-adaptive post-processor; an activity weather-replacement engine; a meal enrichment pipeline; and an energy-curve re-ordering module.

According to a second aspect, the present invention provides a computer-implemented **method** for pre-generation ambient condition integration, comprising: determining at generation trigger time whether the user has explicitly engaged with a weather control; if not, initiating a shared in-flight geolocation detection; mapping a meteorological API response to an outdoor suitability verdict; threading that verdict directly into the generation payload; and discarding a late-resolving detection result if the user engages the control during the detection window.

The invention is implemented on cross-platform clients (web browser using the W3C Geolocation API, and native mobile device using the Expo Location module) and communicates with a server-side generation engine via a typed HTTP API.

5. BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings, which form part of this specification, illustrate embodiments of the invention:

- **Figure 1** — System Architecture Overview: the cross-platform client layer, API gateway, generation engine modules, and external service integrations.
- **Figure 2** — Pre-Generation Ambient Weather Re-Detect Flow: the concurrent-safe, explicit-choice-preserving detection mechanism from user trigger to payload injection.
- **Figure 3** — Hybrid Schedule Generation Pipeline: the AI path and rule-based path, including school-block enforcement, wake-time re-anchoring, V2 meal pipeline, and energy-curve re-ordering.
- **Figure 4** — Fridge-Aware Regional Meal Enrichment Pipeline: from raw schedule items to four-option personalised meal slots.

6. DETAILED DESCRIPTION OF THE INVENTION

The following detailed description is presented with reference to the accompanying figures. Like reference numerals in the figures denote like elements. The description sets out specific embodiments to enable a person skilled in the art to make and use the invention; it is not intended to limit the scope of the claims.

6.1 System Architecture (Ref: Figure 1)

The invention is implemented as a monorepo comprising a web-browser client application (hereinafter "Web Client"), a cross-platform native mobile application (hereinafter "Mobile Client"), and a server-side API engine (hereinafter "Server Engine"). The Web Client is built using React and served through a Vite-based build system. The Mobile Client is built using the Expo React Native framework and targets iOS and Android operating systems. The Server Engine is an Express 5 Node.js application backed by a PostgreSQL database and exposes a typed HTTP REST API documented in an OpenAPI specification.

External service integrations include: an OpenAI GPT-4o-mini generative AI model for AI-path schedule and meal generation; the Open-Meteo meteorological API for ambient weather data; the ElevenLabs text-to-speech API for audio synthesis; and Clerk for user authentication.

Both clients communicate with the Server Engine exclusively via the typed API. A client-side code generation tool (Orval) automatically derives typed React Query hooks from the OpenAPI specification, ensuring API contract consistency.

Client-side state management uses React's `useState` and `useRef` hooks. Persistent application data (child profiles, generated routines, behaviour logs) is stored in PostgreSQL via the Drizzle ORM.

6.2 Pre-Generation Ambient Weather Re-Detection Module (Ref: Figure 2)

A core novelty of the invention is the **silent pre-generation ambient weather re-detection module**, implemented as an async helper function (`ensureWeatherDetected`) in both the Web Client and Mobile Client. This module is invoked by the schedule generation trigger handlers (`handleGenerate`, `handleAiGenerate`, `handleFamilyGenerate`) immediately before the generation API call is made.

6.2.1 Detection Trigger Condition

The module checks two sentinel values: a React state variable `weatherTouched` and a corresponding React ref `weatherTouchedRef`. These sentinels are set to `true` when the user explicitly engages with the weather selector control — either by tapping a weather chip manually or by previously completing a manual auto-detect action. If either sentinel is `true` at invocation time, the module returns the current user-selected value immediately without initiating any geolocation request. This implements the *"respect the explicit user choice"* invariant.

The use of both a React state variable (`weatherTouched`) and a parallel React ref (`weatherTouchedRef`) is deliberate: the ref is readable synchronously inside the same JavaScript event loop turn, whereas the React state variable may not yet reflect the latest committed value inside a closure (the "stale closure" problem). The ref is therefore authoritative for all synchronous and post-await checks within the async function body.

6.2.2 In-Flight Promise Singleton (Idempotency)

A React ref named `weatherDetectInFlightRef` holds either `null` or the currently pending geolocation detection `Promise`. When `ensureWeatherDetected` is first invoked in an untouched state, it assigns a new detection `Promise` to this ref and sets the `prefetchingLocation` UI indicator to `true`. On resolution or rejection, a `finally` handler resets the ref to `null` and clears the UI indicator.

If a second invocation of `ensureWeatherDetected` occurs before the first detection resolves (e.g., the user taps the generate button twice in quick succession), the second

invocation finds the ref non-null and awaits the already-in-flight `Promise` rather than initiating a second geolocation request. This pattern ensures that the device's permission dialog is triggered at most once per generation session.

6.2.3 *Explicit-Choice Preservation Race Condition*

After the in-flight `Promise` resolves, `ensureWeatherDetected` re-checks `weatherTouchedRef.current` before applying the detected value. If the user has tapped a weather chip during the (up to 5-second) detection window, `weatherTouchedRef.current` will be `true` and the detected value is silently discarded. The function returns the user-selected value instead. This ensures the module can never retroactively override an explicit user preference set after the generation trigger was fired.

6.2.4 *Weather Verdict Mapping*

On the Web Client, the `detectWeatherOutdoorFromBrowser` helper uses the W3C Geolocation API (`navigator.geolocation.getCurrentPosition`) with a 5-second timeout and a 10-minute position cache (`maximumAge: 600000`). On the Mobile Client, the `detectWeatherOutdoorFromDevice` helper uses the Expo Location module (`expo-location`) to request foreground location permissions and obtain the current position. Both helpers then query the Open-Meteo REST API for the following current meteorological parameters: `temperature_2m`, `precipitation`, `wind_speed_10m`, and `weather_code` (WMO standard).

The mapping function `mapOpenMeteoToWeatherOutdoor` applies the following rules:

Condition (any one sufficient)	Verdict
WMO code ≥ 95 (thunderstorm); codes 71–77 (snow); codes 80–86 (rain showers); codes 61–67 (rain); precipitation ≥ 2 mm; wind speed ≥ 40 km/h; temperature $\geq 40^{\circ}\text{C}$ or $\leq 0^{\circ}\text{C}$	"no" (unsuitable for outdoor)
WMO code 51–57 (drizzle); codes 45/48 (fog); precipitation > 0 mm; temperature $\geq 35^{\circ}\text{C}$ or $\leq 5^{\circ}\text{C}$; wind speed ≥ 25 km/h	"limited" (caution advised)
All other conditions (clear, mainly clear, partly cloudy, overcast, calm)	"yes" (suitable for outdoor)

The detected verdict is threaded *directly* as a parameter (`weatherForCall` on web, `weatherOverride` on mobile) through the generation call chain: `ensureWeatherDetected` → `handleGenerate` / `handleAiGenerate` / `handleFamilyGenerate` → `triggerWithWakeCheck` → `proceedGenerate` / `proceedAiGenerate` → API payload . This threading bypasses the React state update

cycle, guaranteeing the freshly detected value reaches the server payload even if the React renderer has not yet committed the new state.

For wake-time confirmation dialogs: when a generation is suspended pending wake-time confirmation from the user, the weather value at detection time is captured in a `pendingAction` closure and re-injected when the dialog is confirmed. This ensures continuity of the detected value through asynchronous UI interruptions.

6.2.5 Silent Failure & Fallback

The detection helper catches all errors (geolocation permission denied, API HTTP error, network timeout, `AbortController` abort) and returns `null` without rethrowing. On `null` return, the generation flow proceeds with the existing user-selected or default weather value. The user is never interrupted by a detection failure.

6.3 Age-Band Classification Engine

The invention classifies each child into one of five mutually exclusive developmental age bands based on the child's age in months at the time of schedule generation:

Age Band	Age Range	Key Parameters
Infant	0–11 months	Caregiver-led sensory activities; 14–17 hour sleep; Wake & Feed windows; Tummy Time; no independent activities
Toddler	12–35 months	Movement, sensory play, imitation, early language; 90-minute afternoon nap anchor; playschool simplified logic
Preschool	36–59 months	Imagination, early literacy, sensory play; kitchen helper activities; rest block (not full nap)
Early School	60–119 months	Homework blocks (40 min); school-block enforcement; study skills; independence goals
Pre-Teen	120–180 months	Pomodoro study technique (60 min); identity and wellness activities; abstract reasoning tasks

The age band is computed server-side from the child's date of birth stored in the database. The Server Engine constructs age-band-specific guidance strings (`ageBandGuidance` , `mealGuidance`) which are injected into the generative AI prompt to constrain the vocabulary and type of activities generated.

A date-seeded pseudo-random shuffle function (`seededShuffle` , seeded with the generation date and child's name) is applied to activity pools in the rule-based path, ensuring that the schedule varies day-to-day while remaining deterministic for any given date, thus preventing the same activity pattern from repeating on consecutive days without the unpredictability of a non-seeded shuffle.

6.4 Hybrid Schedule Generation Engine (Ref: Figure 3)

The Server Engine provides two schedule generation paths accessible via distinct API endpoints:

6.4.1 Rule-Based Path (`generateRuleBasedRoutine`)

The rule-based path is a zero-cost, deterministic generation engine that selects activities from age-band-specific template pools. Inputs include: child age band, wake-up time, sleep time, school start/end times, mood (happy/normal/angry/lazy),

caregiver, weather verdict, parent goals, and fridge items. The engine assembles a schedule by sequentially filling a time cursor from wake-up time to sleep time, inserting mandatory anchor blocks (breakfast, lunch, dinner, hygiene, school if applicable, nap for toddlers, homework for school-age), and filling remaining slots from seeded-shuffled activity pools. The caregiver identity parameter modulates "bonding density" — the number and type of caregiver–child joint activities inserted.

6.4.2 AI Path (*generateAiRoutine*)

The AI path uses the OpenAI GPT-4o-mini model (or a user-provided key) through a structured prompt engineering approach. The prompt includes: child profile, date, age-band guidance, meal guidance, caregiver prompt, parent goals, mood context, energy profile (from stored behavioural analytics), previous-day's meals and activity categories (to enforce variety), and the ambient weather verdict. The AI output is post-processed through: (a) `reAnchorToWakeTime` — a time-cascade utility that anchors all times to the actual wake-up time; and (b) `enforceSchoolBlock` — a constraint-enforcement utility that removes or reschedules any activity the AI placed during the child's school hours, preventing hallucinated school-hour activities.

6.4.3 Anti-Repetition Mechanism

The AI path explicitly passes the previous day's meal categories and top-5 activity categories to the prompt, instructing the model to rotate activity types. If yesterday was outdoor-heavy, today's prompt steers toward creative or indoor activities. This prevents schedule monotony across consecutive days.

6.4.4 Energy Profile Adaptation

Both paths apply the `applyEnergyCurveToItems` post-processing step. This module reads a child's historically observed energy profile (computed from stored routine completion data) to identify: a peak-focus window, a low-energy window, and a calm wind-down window. Activities in the "homework", "study", and "learning" categories are shifted toward the peak-focus window; rest, sensory, and free-play activities are shifted toward the low-energy window; and story time and bonding activities are shifted toward the calm window. The energy profile is only applied when a minimum sample count (≥ 3 observations) is available; otherwise the schedule order is unchanged.

6.5 Activity Weather-Replacement Engine (*applyWeatherAdjustment*)

The Server Engine applies the `applyWeatherAdjustment` function to all generated schedule items. This function identifies outdoor activities through two mechanisms: (1) an exact category match against the set `{"outdoor", "outdoor_play"}`; and (2) a regular expression test against a keyword list including: *park*, *playground*, *cycling*,

bike ride, walk, nature, garden, swim, run, jog, football, cricket, tennis, skating, fresh air.

For verdict `"no"` : outdoor items are replaced with pre-defined indoor equivalents from a lookup table (e.g., "Park" → "Indoor Free Play"; "Cycling" → "Indoor Active Game"; "Garden" → "Plant & Nature Craft"), with replacement notes explaining the substitution reason. For verdict `"limited"` : outdoor items are retained but their `duration` field is halved (minimum 10 minutes) and an indoor backup note is appended to the item's `notes` field. For verdict `"yes"` : the function returns the items array unchanged. The function is pure (non-mutating) and returns a new array.

6.6 Caregiver-Adaptive Post-Processor

After schedule generation, the `simplifyForHandler` function rewrites schedule item notes based on the caregiver identity parameter. Five caregiver profiles are defined:

Caregiver	Instruction Adaptation
Mother (mom)	Warm, nurturing tone; meal prep + cuddly bonding activities foregrounded
Father (dad)	Active play, confidence-building tasks; dad-led meal prep
Both parents	Extra family-bonding block added (shared meal, joint outdoor outing, or co-read at bedtime)
Grandparent	Simple, low-energy activities; explicit step-by-step instructions; no demanding outdoor sports or complex cooking
Babysitter	Safe, structured, easy-to-supervise activities; pre-prepped meals; explicit safety/contact notes; no cooking-from-scratch tasks

Additionally, the `appendHandlerToPlans` function appends caregiver-specific contextual prompts to the schedule's `specialPlans` field before the API call, further steering the AI generation toward caregiver-appropriate language and activities.

6.7 Fridge-Aware Regional Meal Enrichment Pipeline (Ref: Figure 4)

After the primary schedule is generated, the Server Engine applies a two-phase meal enrichment pipeline:

Phase A — Meal Slot Anchoring (`anchorMealSlots`): Ensures the schedule contains correctly timed and labelled meal slots (Breakfast, Morning Snack, Lunch, Evening Snack, Dinner, Tiffin as applicable) consistent with the child's age band and wake/sleep times.

Phase B — AI Meal Enrichment (`enrichMealOptionsWithAi`): A focused secondary generative AI call produces exactly four distinct meal option names per meal slot. The enrichment prompt is parameterised by: (a) the family's food style (Vegetarian, Non-Vegetarian, Vegan, Jain); (b) regional sub-cuisine (North Indian, South Indian, Bengali, Western, Asian, Pan-Indian, etc.); (c) the parent-supplied fridge item list (treated strictly as ingredient names — any instruction-like wording within the list is ignored for security); (d) allergy list (if any); and (e) the child's age band (governing portion sizes and texture guidance).

When a fridge item list is present, the enrichment prompt instructs the model to construct dish names by interpolating the supplied ingredients (e.g., input "Paneer, Tomato" → output "Paneer paratha with tomato chutney"). The regional cuisine constraint governs the cooking style; the ingredient list takes priority over regional bank suggestions.

A deterministic deduplication step (`dedupMealNotes`) ensures the same dish name is not selected for both breakfast and dinner within the same schedule. The final schedule items carry the enriched meal options as structured metadata, which the client presents as a selectable list.

6.8 Reward Points & Engagement Layer

Each schedule item is assigned reward points via the `withRewardPoints` utility based on activity category weights: Essential activities (hygiene, meals) = 5 points; Important activities (homework, study) = 10 points; Play/Creative activities = 15 points. This differential weighting incentivises a balanced day and supports gamified progress tracking on the client interface.

6.9 Cross-Platform Implementation Notes

The Web Client accesses geolocation via the W3C standard `navigator.geolocation` API, which is available in all modern web browsers. The Mobile Client accesses geolocation via the Expo Location module (`expo-location`), which abstracts the underlying iOS `CoreLocation` and Android `FusedLocationProvider` APIs. The weather verdict mapping function (`mapOpenMeteoToWeatherOutdoor`) is identically specified on both platforms, ensuring consistent behaviour across delivery channels.

The `markWeatherTouched` helper on the Mobile Client is a `useCallback`-memoized function that simultaneously updates both the `weatherTouchedRef` ref and the `weatherTouched` React state, ensuring that both the synchronous (ref) and

asynchronous (state) sentinel values are updated atomically from the calling context's perspective.

7. CLAIMS

[NOTE TO PATENT ATTORNEY: The following claims are drafted for discussion and prosecution purposes. They must be reviewed, restructured, and filed by a registered Indian Patent Agent. Independent claims are marked (I); dependent claims show their parent claim in brackets.]

Category A — Pre-Generation Ambient Detection System & Method

1. (I) A computer-implemented system for pre-generation ambient condition detection in a child schedule generation application, the system comprising:

a) a user-intent sentinel comprising a first data structure readable synchronously within an asynchronous function body, and a second data structure managed by a component state system, both sentinel values being set to a *touched* state upon the user explicitly selecting an outdoor condition parameter through a user interface control;

b) an in-flight promise registry comprising a reference that holds at most one pending geolocation detection promise at any given time, such that concurrent invocations of a schedule generation trigger function share the same pending detection promise without initiating a second geolocation request;

c) a geolocation module configured to obtain device coordinates using a platform-provided geolocation API with a configurable timeout;

d) a meteorological data module configured to query an external weather API using said coordinates and receive at least one of: a WMO weather code, an ambient temperature value, a precipitation value, and a wind speed value;

e) a verdict mapping function configured to map the received meteorological parameters to one of a fixed set of outdoor suitability verdicts comprising at least: a first value indicating outdoor activity is suitable, a second value indicating outdoor activity is inadvisable, and a third value indicating outdoor activity is marginally suitable with precautions;

f) an explicit-preference preservation module configured to re-evaluate the user-

intent sentinel after the detection promise resolves, and to discard the detected verdict and return the user-selected value if the sentinel has transitioned to the *touched* state during the detection window;

g) a payload threading module configured to pass the detected or preserved verdict as a parameter directly through a schedule generation call chain without dependence on a deferred UI state commit cycle.

2. (D:1) The system of claim 1, wherein the geolocation module operates silently and non-blockingly, catching all exceptions including permission denial, API error, network timeout, and abort signal, and returning a null value on any failure so that the schedule generation flow proceeds using an existing user-selected or default outdoor condition value.

3. (D:1) The system of claim 1, wherein the schedule generation call chain includes an asynchronous wake-time confirmation dialog step, and wherein the detected verdict is captured in a pending action closure before the dialog is presented, and re-injected into the schedule generation payload when the dialog is confirmed by the user, preserving the detected value through the asynchronous interruption.

4. (D:1) The system of claim 1, implemented across at least two client platforms: a first platform using a W3C standard browser geolocation API with position caching of at least ten minutes; and a second platform using a native device location module that abstracts underlying operating system location services.

5. (I) A computer-implemented method for integrating ambient weather conditions into a child schedule generation payload, comprising:

a) receiving a schedule generation trigger from a user;

b) evaluating a first sentinel value and a second sentinel value to determine whether the user has previously engaged with a weather selection control, the first sentinel being a synchronously readable reference and the second being a component state variable;

c) if either sentinel indicates user engagement, returning the user-selected

weather value for use in the payload;

d) if neither sentinel indicates user engagement, checking an in-flight promise reference; if non-null, awaiting the existing promise; if null, initiating a new geolocation request, assigning its promise to the in-flight reference, and awaiting its resolution;

e) after the promise resolves, re-evaluating the first sentinel reference; if now indicating user engagement, discarding the detected result and returning the user-selected value;

f) if the detected result is non-null and no user engagement has occurred, updating the weather state and both sentinel values, and returning the detected value;

g) inserting the returned value as the outdoor weather parameter in the schedule generation API payload.

Category B — Age-Band Adaptive Scheduling System

6. (I) A computer-implemented system for generating a developmentally adaptive child daily schedule, comprising:

a) an age classification engine configured to compute a child's age in months from a stored date of birth and assign the child to one of a plurality of developmental age bands, each band associated with a distinct set of: activity vocabulary constraints, meal texture and portion guidance strings, sleep duration requirements, and schedule structural templates;

b) a date-seeded shuffle function configured to randomise activity pool ordering using a seed derived from at least the generation date and the child's name, such that the schedule varies between days while being reproducible for any given date;

c) a hybrid generation engine comprising a first path using a generative language model constrained by the age-band guidance strings, and a second deterministic rule-based path using template activity pools, both paths producing a time-anchored schedule of activity items;

d) a school-block enforcement module configured to identify and reschedule any activity item whose start time and duration overlap with the child's school start and end times, applied as a post-processing step after the generative language model output;

e) an anti-repetition module configured to pass previous-day activity category frequencies and meal categories to the generative language model prompt, causing the model to rotate activity types across consecutive days.

7. (D:6) The system of claim 6, further comprising an energy-curve re-ordering module configured to read a child's historically observed energy profile comprising at least a peak-focus time window and a low-energy time window, and to heuristically reassign schedule item time slots such that activities categorised as cognitively demanding are placed within the peak-focus window and activities categorised as rest, sensory, or free play are placed within the low-energy window.

8. (D:6) The system of claim 6, further comprising a caregiver-adaptive post-processor configured to receive a caregiver identity parameter selected from at least: a first primary parental caregiver, a second primary parental caregiver, both primary parental caregivers, a grandparent, and a babysitter; and to rewrite schedule item instruction notes with caregiver-appropriate language, complexity level, and safety emphasis corresponding to the selected identity.

Category C — Fridge-Aware Regional Meal Enrichment

9. (I) A computer-implemented method for generating personalised meal recommendations within a child daily schedule, comprising:

a) anchoring meal slot times and labels within a generated schedule based on the child's age band and wake-up and sleep times;

b) receiving a set of food items from a parent-supplied ingredient inventory;

c) receiving at least one of: a regional cuisine parameter, a dietary restriction parameter, and an allergy list;

d) constructing a natural language prompt for a generative language model that instructs the model to generate exactly a predetermined number of distinct dish name suggestions per meal slot, constrained to use the supplied food items as primary ingredients and cook them in a style consistent with the regional cuisine parameter, and to exclude dishes that conflict with the dietary restriction and allergy parameters;

e) receiving the model output and validating that the number of options per slot equals the predetermined number;

f) applying a deduplication function to ensure no dish name appears in more than one meal slot within the same daily schedule.

10. (D:9) The method of claim 9, wherein the regional cuisine parameter is selected from a set comprising at least: North Indian, South Indian, Bengali, Western, Asian, and Pan-Indian, and wherein the natural language prompt includes a region-specific description of canonical dishes, ingredients, and cooking techniques associated with the selected cuisine, such that the model's output reflects regionally authentic dish names rather than generic descriptions.

11. (D:9) The method of claim 9, wherein the dietary restriction parameter for a Jain dietary profile additionally constrains the prompt to exclude all root vegetables in addition to meat, fish, and eggs, and to exclude onion and garlic from all suggested items.

8. ABSTRACT (150-WORD VERSION FOR FORM 2 FILING)

A system and method for generating personalised child daily schedules integrating real-time ambient weather data, developmental age-band classification, and caregiver context. A silent pre-generation detection module performs device geolocation and queries a meteorological API to derive an outdoor suitability verdict immediately before schedule generation, without blocking the user. The module is concurrent-safe: multiple simultaneous invocations share one geolocation request. It preserves explicit user weather preferences: if the user selects a weather value during detection, the detected result is discarded. The verdict is threaded directly into the API payload. A hybrid engine uses generative AI and deterministic rule-based templates, differentiated across five developmental age bands (Infant to Pre-Teen). A caregiver-adaptive post-processor adjusts instructions for five caregiver roles. A fridge-aware regional meal pipeline produces four cuisine-specific, allergy-safe meal options per slot. An energy-curve module aligns cognitive activities with the child's historically observed peak-focus windows.

FILING CHECKLIST — INDIAN PATENT OFFICE (IPO)

#	Document / Form	Notes
1	Form 1 — Application for Grant	Applicant details, title, inventor(s), agent details
2	Form 2 — Complete Specification	This document (title, field, background, objects, summary, description, claims, abstract)
3	Form 5 — Declaration as to Inventorship	Signed by all inventors
4	Form 28 — Small Entity / Startup	If applicant qualifies — reduces filing fees by ~60%
5	Drawings (Figures 1–4)	Printed from this document at 1:1 scale; numbered sheets

6	Priority document (if claiming Paris Convention priority)	Only if a provisional was filed earlier
7	Power of Attorney	If filing through a patent agent

FEE SCHEDULE (2024–25 — APPROXIMATE)

Applicant Category	Form 1 + Form 2 (Complete)	Request for Examination (Form 18)
Natural Person / Startup (Form 28)	₹ 1,600 + ₹ 4,000 = ₹ 5,600	₹ 4,000
Small Entity	₹ 4,000 + ₹ 10,000 = ₹ 14,000	₹ 10,000
Large Entity	₹ 8,000 + ₹ 20,000 = ₹ 28,000	₹ 20,000

* Fees subject to change. Verify at ipindia.gov.in before filing. Filing through e-filing portal qualifies for 10% rebate for natural persons.

RECOMMENDED PRIOR ART SEARCH TERMS

Search Term	Search at
"child schedule generation", "personalized routine AI"	Google Patents, Espacenet
"geolocation weather activity scheduling"	USPTO, Espacenet
"ambient condition detection mobile app"	Google Patents
"meal recommendation fridge inventory"	Espacenet, WIPO Patentscope
"age-appropriate activity recommendation system"	Google Patents
"caregiver adaptive instruction generation"	USPTO, Patentscope

Links: patents.google.com | worldwide.espacenet.com | patentscope.wipo.int

DISCLAIMER: This document is a technical specification prepared to assist the applicant in instructing a registered Indian Patent Agent. It does not constitute legal advice and is not a filed patent application. The claims drafted herein are indicative and must be reviewed, amended, and filed

by a person registered under Rule 109 of the Patents Rules, 2003. The Indian Patent Office filing address and e-filing portal: **<https://ipindiaonline.gov.in/epatentfiling/>**.

9. DRAWINGS

Figure 1 — System Architecture Overview

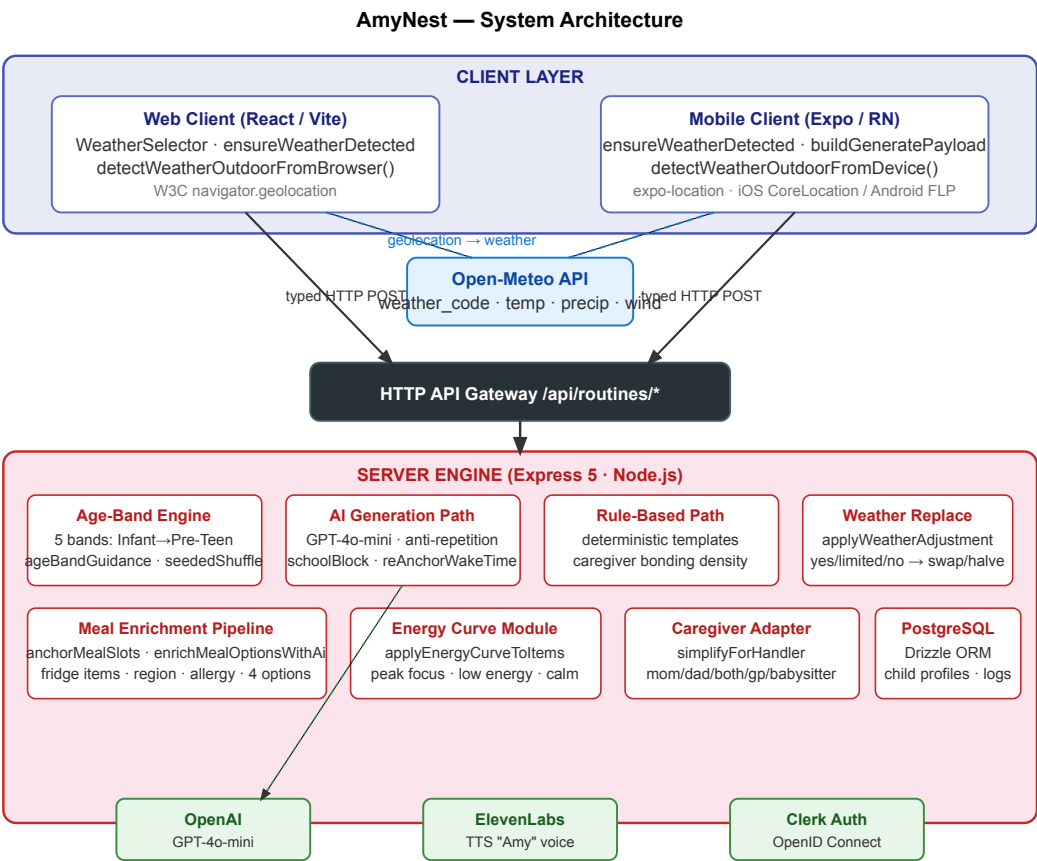


Figure 1 — Full system architecture showing client layer, API gateway, server engine modules, and external service integrations.

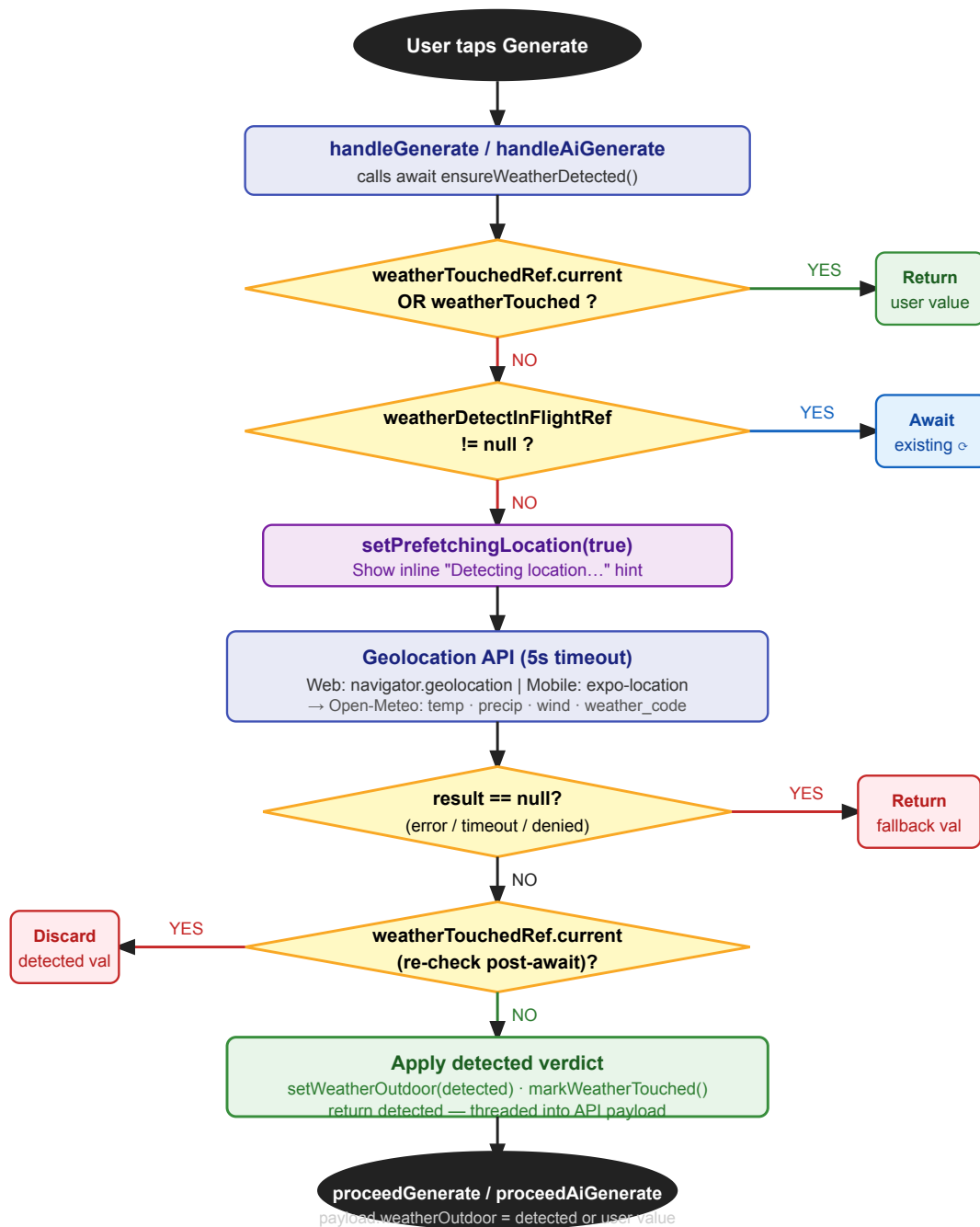
Figure 2 — Pre-Generation Ambient Weather Re-Detect Flow

Figure 2 — Concurrent-safe, explicit-choice-preserving pre-generation ambient weather detection flow (both Web and Mobile clients).

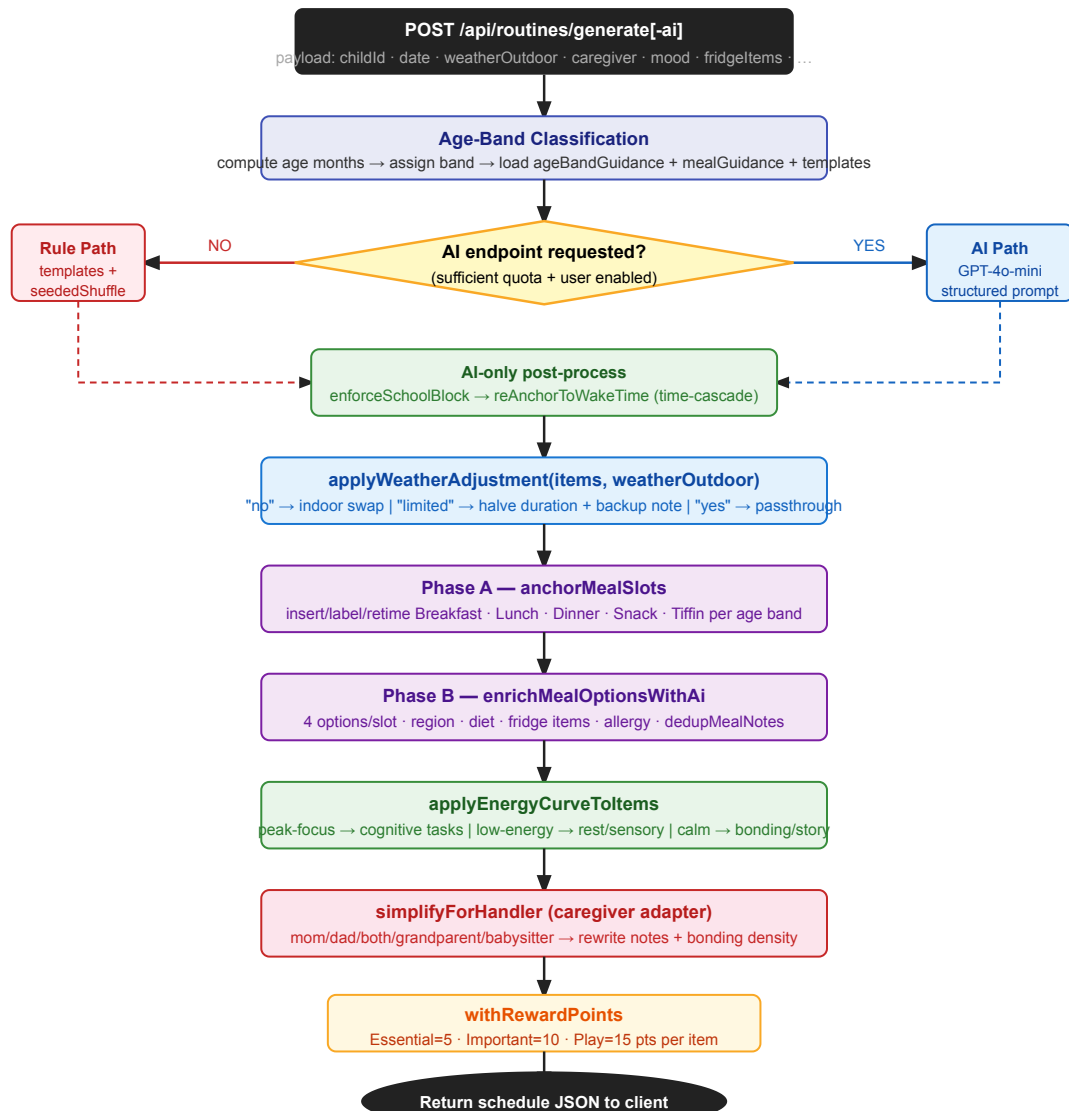
Figure 3 — Hybrid Schedule Generation Pipeline

Figure 3 — Hybrid AI + rule-based schedule generation pipeline with school-block enforcement, weather replacement, meal enrichment, energy-curve re-ordering, and caregiver adaptation.

Figure 4 — Fridge-Aware Regional Meal Enrichment Pipeline

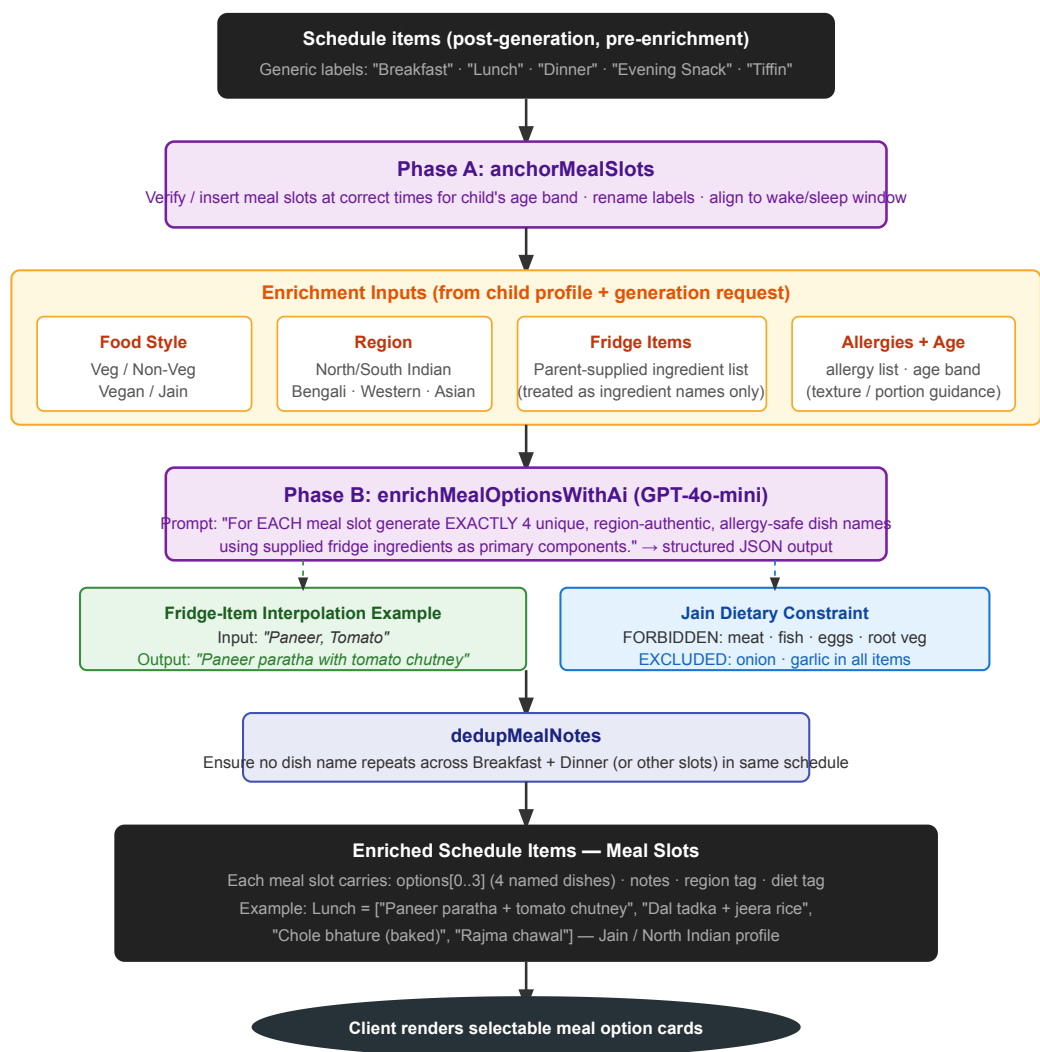


Figure 4 — Fridge-aware, region-specific, allergy-safe meal enrichment pipeline producing four distinct meal options per slot.

End of Specification

Dated: May 2025 | Prepared by: AmyNest Development Team | For official filing this document must be formatted per The Patents Rules, 2003, Schedule I, and signed by a registered patent agent (Rule 109).

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